

Wesam Aldiab

✉ Wissamdiab122@gmail.com | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

📍 Adelaide, SA - ☎ +61 424 558 047

Education

Bachelor of Engineering (Honours) (Software) | Adelaide University

Jul 2024 – Oct 2028

Skills

Programming: { C++ | Ruby | HTML/CSS | MATLAB | HDL (Nand2Tetris) }
Framework / Tools: { Ruby on Rails | Git | Excel }
Core CS: { [Data Structures & Algorithms](#) }
Software Design: { OOP (SOLID) | [Design Patterns](#) | UML (Class, Sequence, Use Case) }
Languages (spoken): { English | Arabic }

Projects

- **[Nonogram Solver \(C++\)](#)** **Jan 2026 – Feb 2026**
 - Command-line solver that loads puzzles from a plain-text format and prints the solved grid (or best partial state).
 - Two-stage approach: fast trivial-constraints pass, then CSP solver with domain propagation and AC-3 style queue.
 - DFS backtracking with a "most constrained cell" heuristic; line propagation uses an automaton-style state model for pruning.
- **[Digital Logic Systems \(HDL, VM, Compiler\)](#)** **Jul 2025 – Oct 2025**
 - Implemented an assembler, virtual machine translator, and compiler for a simple object-oriented language, demonstrating code generation, stack-based VM execution, and runtime/OS abstractions.
- **[Pong Game Physics Engine C++](#)** **Jul 2025 – Oct 2025**
 - Co-built a 2-player Pong game using a teammate's custom framework (not redistributable) for windowing/input/rendering.
 - Implemented time-of-impact collision detection and response to prevent tunnelling; handled multiple collisions per tick (e.g., corner impacts).
 - Built ball-wall and ball-paddle interactions with manual vector maths (normals/reflection) and clean, modular entity updates via virtual functions.
 - Added hot-reloadable JSON config via file monitoring + auto save/load, plus FPS/debug overlay and reset/exit controls

Community Involvement

- **[Ravi's Study Program \(RSP\)](#)** **Dec 2025 – Feb 2026**

RSP is a 13-week, no-cost, volunteer-led summer bootcamp (Nov–Feb) focused on data structures & algorithms and interview-style problem solving.
- **[Competitive Programming \(ACPC × RSP\)](#)** **1 Feb 2026**

Competed in the Adelaide Competitive Programming Club × Ravi's Study Program contest team solved under timed conditions.